

Question 1 | Correct | Mark 10.00 out of 10.00**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3. Write any efficient algorithm to find the possible ways.

Example 1:

Input: 6

Output: 6

Explanation: There are 6 ways to 6 represent number with 1 and 3

$1+1+1+1+1+1$

$3+3$

$1+1+1+3$

$1+1+3+1$

$1+3+1+1$

$3+1+1+1$

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2  int main() {
3      int n;
4      scanf("%d", &n);
5      long long dp[n + 1];
6      dp[0] = 1;
7      for (int i = 1; i <= n; i++) {
8          dp[i] = dp[i - 1];
9          if (i >= 3) {
10             dp[i] += dp[i - 3];
11         }
12     }
13     printf("%lld", dp[n]);
14     return 0;
15 }
```

	Input	Expected	Got	
✓	6	6	6	✓
✓	25	8641	8641	✓